

- 1 Fig. 1.1 shows the variation of velocity v with time t for a fuel-propelled model rocket travelling upwards.

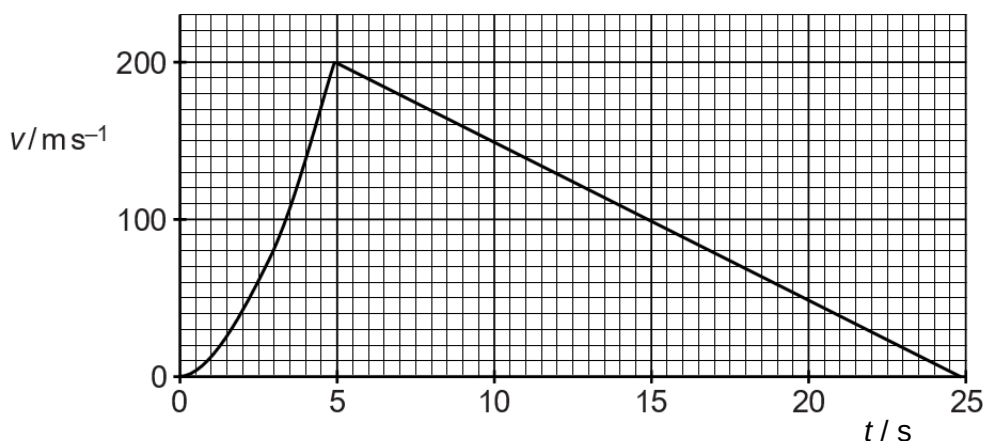


Fig. 1.1

- (a) (i) Define *acceleration*.

It is the rate of change of velocity [B1][1]

- (ii) State the time interval when the rocket has the greatest acceleration.

between $t = 3.5$ to 4.0 s to $t = 5.0$ s [1] [B1]

- (iii) Explain why the acceleration of the rocket increases between $t = 0$ to $t = 5.0$ s.

[Intended ans] Fuel is burnt and mass gradually decreases, so with a constant thrust force, net force increases causing acceleration to increase [B1]

.....[1]

- (b) (i) Suggest a reason for the abrupt change in the motion of the rocket at $t = 5.0$ s.

All fuel has been burnt / No more thrust [B1]

.....[1]

- (ii) Explain how it can be deduced from Fig. 1.1 that air resistance has a negligible effect on the motion of the rocket.

As the rocket continues to rise, it experiences constant deceleration between $t = 5.0$ s to $t = 25$ s [B1]

numerically equal to 10 m s^{-2} (or value of g stated) [B1]

OR reverse argument: If air resistance were not negligible, then deceleration would be $> g$ and not constant as indicated in Fig. 1.1. [2]

4

(c) Use Fig. 1.1 to determine the maximum height reached by the rocket.

max height = area under graph from $t = 0$ to $t = 25$ s

$$= (73 \text{ squares} \times 0.5 \times 10) + \frac{1}{2}(20)(200) \quad [\text{M1}]$$

$$= 365 + 2000$$

$$= 2365 \text{ m} \quad [\text{A1}]$$

height = 2350 to 2380 m [2]

[Total: 8]

3 (a) State the principle of moments